

Sultanate of Oman

Nama Water Services Company SAOC (NWS / OWWSC)

Consultancy Services for Design and Supervision for STP, Sewer & TE Networks Systems in Ibri

Tender No. T/2719110/2025

Concept Screening Report

Revision R1 — July 2026

Renardet S.A. & Partners Consulting Engineers

Project No. 2621



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1. EXECUTIVE SUMMARY

This report presents a desk-top concept screening of the wastewater collection strategy for Ibri Wilayat, prepared ahead of the project kick-off. Its purpose is to establish, on the basis of the available terrain model, cadastral and road data, and strictly in accordance with the NWS design guidelines (PAM-GUD-203 and PAM-GUD-201), the broad shape of the future sewer system: where the flows are generated, how they can reach the existing Ibri STP site, which areas can be served by gravity, where sewage lifting stations (SLS) are unavoidable, and what the ultimate flows imply for the treatment capacity.

The principal findings are:

- Ibri is fundamentally a gravity catchment. The main agglomeration lies 45 to 90 m above the existing STP site, providing a median available fall of about 3.6 m/km along the natural drainage direction — several times the minimum gradient required for large trunk sewers (0.75 mm/m for DN900 and larger, Table 11 of PAM-GUD-203). A gravity trunk system of approximately 22 km of main trunk and 172 km of trunk branches can serve the entire boundary.
- The serviceable land use comprises 53,503 plots (43,722 residential). Applying the PAM-GUD-201 methodology (164 l/c/d for Adh Dhahirah, non-domestic and governmental ratios, return rates, linear infiltration allowance) the ultimate saturated average flow is approximately 49,700 m³/d, i.e. about 54,700 m³/d with the 10% STP design margin. Since this exceeds the 20,000 m³/d Phase I threshold defined in the Terms of Reference, phasing of the treatment capacity against realistic build-out horizons (2030 / 2055 / ultimate) becomes the central concept decision of the project.
- The study area was partitioned into 36 sewer zones, each a contiguous road-network territory draining to a single outlet on the trunk system. Zone boundaries follow the road fabric and plot density rather than pure terrain watersheds.
- A minimum-gradient depth-accumulation screening (Table 11 gradients, 12 m maximum cover) shows that about 84% of plots reach the STP by gravity. The remaining pockets — mostly wadi-floor villages south-west and north-west of town lying below the STP inlet level —



consolidate into 18 candidate sewage lifting stations after cascading nearby pockets together; 147 minor pockets are left to be absorbed by local re-routing during design.

- The trunk network crosses mapped wadi lines at approximately 134 locations (12 on the main trunk). Crossing design obligations (ductile iron across the crossing plus 15 m either side, 2.0 m minimum cover in soft soil, valve arrangements, MoAFWR approvals and 1:20/1:50/1:100 flood data) are summarised in this report.
- A structured register of data required from NWS and third parties (population and occupancy statistics, water billing and flow-meter records, existing STP/PS records, as-builts, master plan extracts) is provided; several parameters of this screening are provisional pending that data and shall be confirmed before concept design.

All results are screening-grade: they are derived from an unverified terrain model and cadastral data, and are intended to focus the data collection, survey and modelling effort of the concept stage — not to fix any design parameter.



2. INTRODUCTION AND PROJECT BACKGROUND

Nama Water Services (NWS/OWWSC) intends to extend wastewater services to all unserved plots within the operational sewer and treated-effluent (TE) service area of Ibri Wilayat, to verify the RG Master Plan 2018, and to develop the concept, preliminary and detailed designs and tender documents for the wastewater network, the TE network and the sewage treatment capacity, for a design horizon equal to the year of project completion plus 25 years, or ultimate (saturated) conditions (TOR Section 03).

The scope explicitly comprises two parallel piped systems: the wastewater collection system (property connections, laterals, main sewers, trunk mains, lifting/pumping stations and rising mains, modelled in SewerGEMS) and the treated-effluent distribution system (TE mains and customer connections from the STP to TE customers, modelled in WaterGEMS), together with the new STP capacity. For the STP, the Terms of Reference make the preliminary design and EPC tender preparation part of this consultancy only where the Phase I design capacity is below 20,000 m³/d; at greater capacities those stages are subject to separate NWS instruction.

This Concept Screening Report is a preparatory engineering assessment. It precedes the formal data collection, topographical survey, geotechnical investigation and hydraulic modelling of the concept stage, and its purpose is threefold: (i) to apply the NWS design criteria to the available data and establish the feasible shape of the collection system; (ii) to identify the decisions and risks that will drive the concept options; and (iii) to define precisely which data must be obtained before the concept design can be fixed.

2.1. DESIGN BASIS DOCUMENTS

- PAM-GUD-203 — Wastewater Design Guidelines v1.0 (Rev 01). Cited in this report as (p##).
- PAM-GUD-201 — General Design Guidelines v1.0 (Rev 01). Cited as (G1-p##).
- Terms of Reference, Section 03, Tender T/2719110/2025 (cited as TOR).



3. SCOPE OF SERVICES AND DELIVERABLES

The consultancy proceeds in stages: Concept Design and associated studies; Preliminary Design; Detailed Design; Tender Documentation and tendering support; followed by construction supervision. The obligations most relevant to the present screening are summarised below.

3.1. KEY TECHNICAL OBLIGATIONS

Obligation	Requirement	Reference
Coverage	All plots within the project boundary — built, open and under construction — shall be provided with sewer and TE connections.	TOR H.2
Gravity preference	Sewage connection by gravity is preferred; pumping only where gravity is not realistic or is excessively costly, with both options considered.	TOR p10
Topography-led layout	The network layout shall utilise gravity as far as practicable using contour mapping to avoid pumping.	TOR p12
Options	Minimum three options each for the wastewater network, the TE network, and each STP.	TOR p13
Design horizon	Completion + 25 years or ultimate (saturated); model years start / 2030 / 2055 / ultimate at 5-year projection intervals.	TOR p3, p14
Modelling platforms	SewerGEMS (wastewater) and WaterGEMS (TE), with Excel calculation deliverables.	TOR p14
Inverted siphons	To be avoided; permitted only where no feasible alternative exists.	TOR p12
STP Phase I threshold	Preliminary design and EPC tender for the new STP fall within this consultancy only if Phase I capacity is below 20,000 m ³ /d.	TOR p3, p5

3.2. SURVEYS AND INVESTIGATIONS IN THE CONSULTANT SCOPE

The designs shall be based on surveys and investigations executed under this consultancy, not on the information folder, which is provided for information only (TOR H.14):

- Topographical survey of the whole project area, with ground levels, cover levels, inverts and asset data of the existing sewer and TE systems, delivered to NWS as-built/GIS specifications (TOR H.11, Concept deliverables).
- Geotechnical investigation at concept and design stages: soil profile and strength for excavation methods and shoring, pipe bedding and structural design of deep trunk sewers, manholes and pump stations, groundwater levels for dewatering assessment, and chemical aggressivity for material selection (TOR H.12; deliverables 20-22).



- Trial pits — fifty (50) at critical locations proposed by the Consultant and approved by NWS: their purpose is the physical identification of existing underground services (position, depth, size) at road intersections, along major utility corridors and along the expected routes of trunk sewers and rising mains, to secure constructability and clash-free alignments; they also verify groundwater conditions locally (TOR H.13, Utility Surveys clause).
- CCTV and condition surveys of existing assets where rehabilitation or integration is required (per NF EN 13508-2 / GUD-203 Section 13).

3.3. CONCEPT-STAGE DELIVERABLES

The concept design report shall include, inter alia: executive summary; data collection and validation report; as-built and GIS of existing sewer and TE assets; population forecast and flow projections at 5-year intervals; wastewater generation, design flows and flow directions; minimum three network options with hydraulic calculations (SewerGEMS/WaterGEMS); concept designs for lifting/pumping stations wherever required; STP location report and concept design with at least three options per STP; TE and sludge management strategy; tanker discharge and TE filling facilities; emergency lagoons (5-day storage) and excess-TE wadi discharge provisions; life-cycle costs; environmental and social screening; contracting strategy and packaging (TOR 4.1.1-4.1.1.2).



4. DESIGN BASIS AND CRITERIA

This section presents the criteria that governed the screening, with their engineering rationale. All values are taken from the NWS guidelines without modification; where the guidelines defer a value to other sources, this is stated explicitly and the value is carried as an assumption to be confirmed.

4.1. HYDRAULIC DESIGN OF GRAVITY SEWERS

Gravity sewers are designed with the Colebrook-White formula using an operational roughness $k_s = 1.5$ mm for all pipe sizes and materials, or alternatively Manning's formula, computed in NWS-approved software (p24). The roughness value represents an aged pipe in sewage service, including joint and slime effects, rather than the catalogue value of the new pipe.

Two velocity limits bound the design. The minimum self-cleansing velocity of 0.75 m/s at peak flow (0.90 m/s preferred) ensures that sand and organic solids do not settle permanently; where early flows cannot reach it — typically at the heads of the system — the guideline requires a minimum tractive-force (boundary shear) check instead, which recognises that a shallow fast film can transport sediment even at low discharge (p26-27). The maximum velocity of 3.0 m/s protects the pipe invert and structures from abrasion and prevents excessive turbulence, splashing and release of hydrogen sulphide (p27).

At peak flow the proportional depth d/D is limited to 0.65 for pipes up to DN350 and 0.50 above DN350 (p27). The reserve above the flow surface ventilates the sewer, limits septicity, and retains capacity for flows beyond the design estimate; the stricter limit for large sewers reflects the greater consequence of surcharge in trunk mains.

These hydraulic requirements translate into the minimum gradients of Table 11 (p29), reproduced below; the steeper of the Table 11 value and the tractive-force value governs. Sewers shall not be oversized to justify flatter gradients, and uniform gradients are required between manholes (p29).

Sewer diameter (mm)	Minimum gradient (mm/m)
200	5.00
250	3.75
315	2.70
400	2.05



500	1.55
600	1.25
700	1.00
800	0.85
900 and above	0.75

Table — Minimum sewer gradients, PAM-GUD-203 Table 11 (p29)

4.2. DEPTH, COVER AND MANHOLES

The minimum depth to pipe crown is 1.3 m (0.5 m where concrete protection is provided), protecting the pipe from traffic loads and utility conflicts (p33). The recommended maximum cover is approximately 10-12 m: beyond it, excavation, shoring and dewatering costs escalate and maintenance access deteriorates, and the guideline directs the designer to introduce a pumping station rather than deepen the sewer further (p33). This 12 m ceiling, combined with the Table 11 gradients, is the criterion by which the present screening decides where gravity fails.

Manholes are required at every change of gradient, diameter or alignment, at junctions and lateral ends, at maximum spacings of 100 m (DN200-315), 120 m (DN350-900), 150 m (DN1000-1400) and 200 m above DN1400 (p30). Drops above 600 mm require external backdrops, limited to 2 m height (p30). Manholes and pipelines shall not be located in wadis or areas subject to washout (p30, p33).

4.3. TRUNK MAINS

NWS defines trunk mains as gravity pipelines larger than DN800, longer than 1,000 m without connections, immediately upstream of the STP or a main pumping station (p35). Materials above DN600 are GRP, lined reinforced concrete, or profile-wall HDPE (p35). Service corridors range from 2.0 m (up to DN500) to 4.4 m (DN2400) (p32-35).

4.4. PUMPING STATIONS AND FORCE MAINS

Where pumping is unavoidable, stations are sited on hydraulic merit — flooded suction, minimised energy, force-main economics (p38) — with duty/standby pumps each capable of the design flow in small stations (p39). Force mains maintain 0.75 m/s minimum (1.0 m/s where flow is intermittent), 2.5 m/s maximum; gradients not flatter than 1:750, ideally 1:500 rising / 1:300 falling; retention below 30 minutes where achievable; air valves at high points and washouts at low points; access every 500



m; 3.0 m horizontal clearance from water mains, crossing beneath them with 450 mm vertical separation (p50-51).

4.5. WADI CROSSINGS

Crossing design requires hydrological data (wadi profiles, 1:20, 1:50 and 1:100 year floods, bed material, bed-level trends) obtained from CAA and MoAFWR, with MoAFWR approval (G1-p85). Ductile iron pipe with restrained joints is used across the crossing and 15 m beyond each bank; protection follows standard drawing PAM-STD-404; anti-flotation is checked for the empty pipe under flood; minimum cover in soft soil is 2.0 m; isolation and air valves are placed on both banks with a washout at the low point; and no chambers or marker posts are permitted in the wadi bed or embankments (G1-p86). Excess treated effluent discharged to a wadi must meet Class A quality (MD 145/93) and requires Environment Authority approval (p73).

4.6. FLOW ESTIMATION METHODOLOGY

Flows are computed by the PAM-GUD-201 chain: population is derived from plots (plots x average properties per plot x occupancy rate, with the occupancy rate taken from NCSI housing statistics, G1-p58); domestic water consumption for Adh Dhahirah Governorate is 164 l/c/d (IMP 2024 baseline, G1-p60), increased by the governorate ratios of 22% for non-domestic and 14% for governmental consumption (G1-p60); wastewater generation applies return rates of 85% for domestic and tanker consumption and 54% for non-domestic and governmental consumption (G1-p71); a linear infiltration allowance of 720 L/d per km of new sewer is added (G1-p72); the wastewater peaking factor follows the Peltier formula $Pf_{WW} = 1.5 + 1/\sqrt{Q_m}$ with Q_m in L/s, capped at 5.0 (G1-p72); and STP capacity carries an additional 10% operational margin (G1-p73). The design horizon of the networks is completion + 25 years or saturation (TOR), while STPs are designed for phased horizons of at least 15 years (p65).

For the present screening, the occupancy rate has been provisionally taken as 6.0 persons per housing unit with 1 property per plot; both parameters shall be replaced by NCSI figures before concept design, and all flows scale linearly with them.



5. STUDY AREA

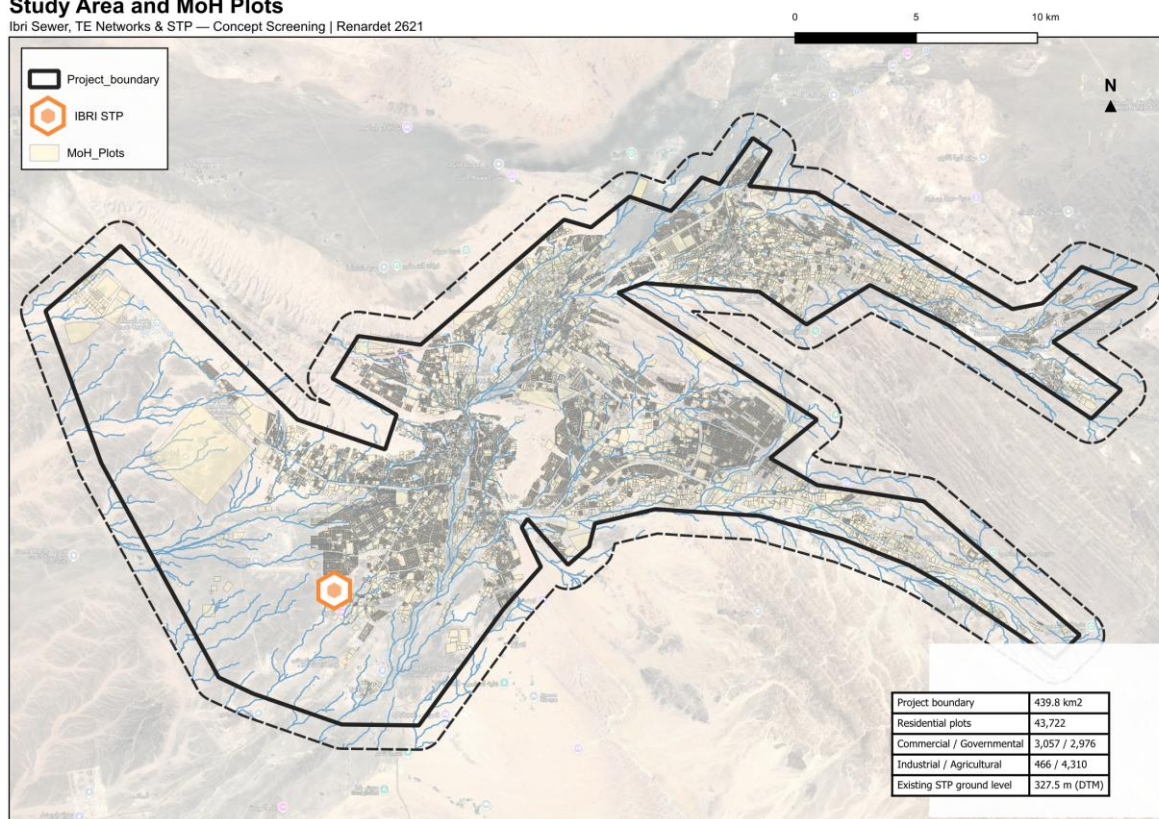
The project boundary covers 439.8 km² of Ibri Wilayat in Adh Dhahirah Governorate. Urban development concentrates in the central agglomeration of Ibri town, extending along the dual-carriageway corridors north-eastwards (Al Araqi and beyond) and south-eastwards, with satellite settlements to the east and south. The existing Ibri STP occupies a low position at the south-western edge of the town at ground level of approximately 327.5 m.

Land use class	Plots	Area (km ²)
Residential	43,722	32.2
Agricultural	4,310	36.8
Governmental	2,976	16.3
Commercial	3,057	2.4
Industrial	466	0.9
Not classified	3,991	7.5

Table — Land use within the project boundary (MoHUP cadastre)

Study Area and MoH Plots

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Map M1 — Study area, MoH plots and existing STP

5.1. EXISTING SEWER AND TE SYSTEM

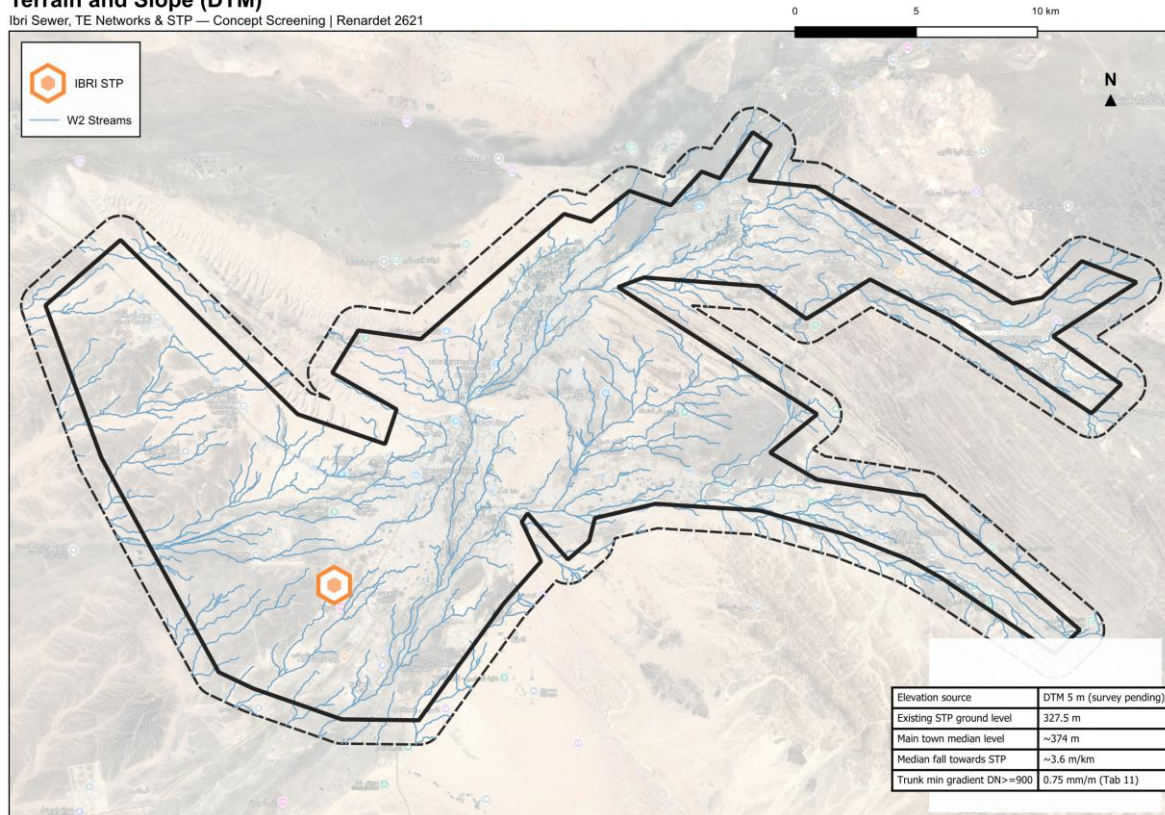
Figure No. 2 of the Terms of Reference (Existing Sewer and TE Systems) shows that a sewer system is already in operation in the north-eastern district of the agglomeration (Al Araqi area), connected to the existing STP by a trunk corridor running south-west through the town. The concept design must therefore (i) verify the route, capacity, condition and hydraulic performance of this existing trunk and its district network against the design-horizon flows, (ii) integrate or redesign it as required, and (iii) decide whether the new trunk system parallels, reuses or replaces the existing corridor. The as-built records, drawings and previous design documentation of this system are first-priority data items (Section 8). Zones overlapping the served district contain existing customers, whose flows are already reaching the STP and must not be double-counted in phasing.

Terrain falls from the jebel foothills in the north and east towards the south-west. Within the built-up area, ground levels range from about 320 m near the STP to about 490 m at the north-eastern settlements. Wadi channels dissect the entire area and control both the gravity drainage pattern and the crossing inventory of any trunk system. Terrain analysis in this report uses the project digital terrain model (5 m); as the topographic survey is ongoing, all levels are indicative.



Terrain and Slope (DTM)

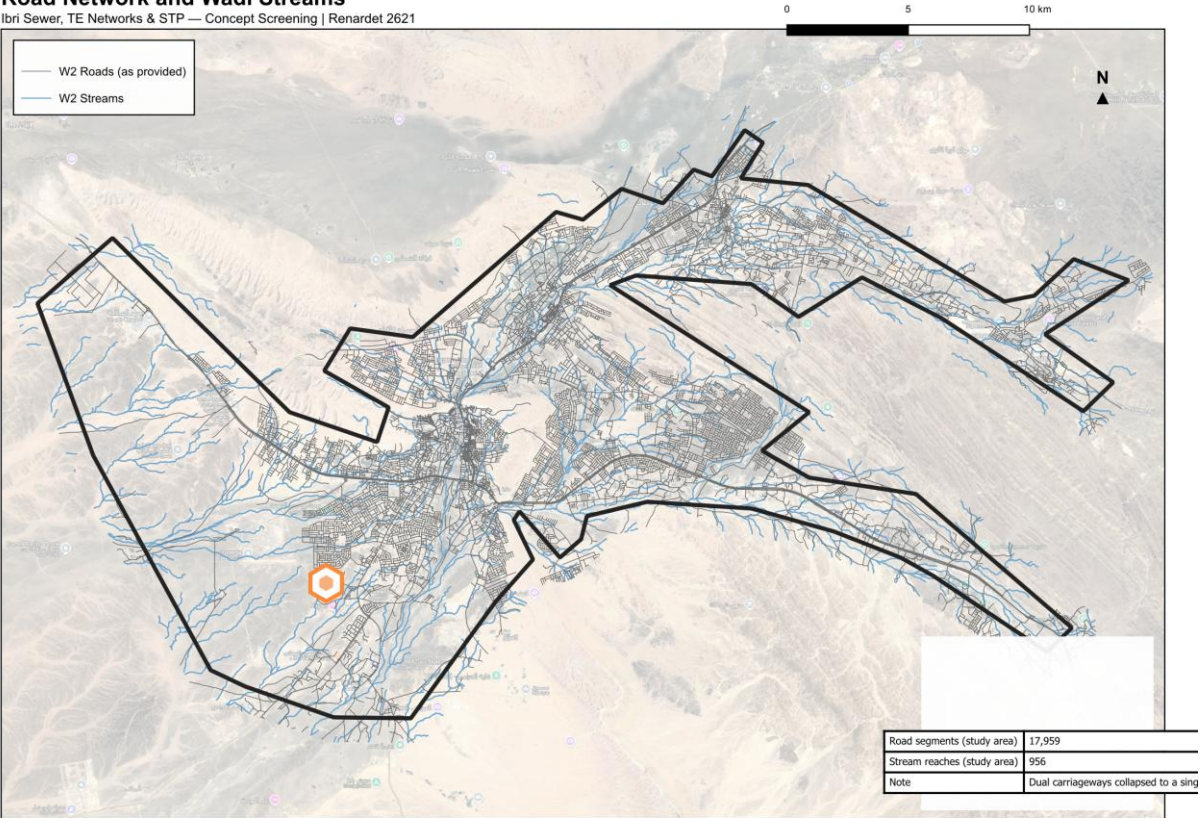
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Map M2 — Terrain and drainage (DTM)

Road Network and Wadi Streams

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Map M3 — Road network and wadi streams



6. SCREENING METHODOLOGY

The screening treats the public road network as the universe of feasible sewer corridors, consistent with the guideline requirement that the secondary network follows the street layout (p32). The road centrelines were converted into a connected network of about 13,400 junction nodes and 17,600 street segments. Where a main road is represented by two parallel carriageway centrelines, the pair was collapsed into a single routing corridor, so that a trunk sewer is routed once along the road reserve rather than once per carriageway.

Every serviceable plot was assigned to its nearest street node, giving each node a demand weight in plots. Ground levels at all nodes were sampled from the DTM. From the existing STP, least-cost routes were computed to every node through the street network; the cost function prefers major-road corridors (which offer wide reserves and direct alignments and avoid dense internal streets) and penalises routes that would force sewage to climb, so that the resulting trunk paths track the natural fall of the terrain along main roads with as few directional changes as practicable.

Sewer zones were then formed as road-network territories: each of the 36 selected outlet points on the trunk system claims the street nodes closer to it (through the network) than to any other outlet, weighted by plot density. Zone boundaries therefore follow the road fabric and the actual distribution of plots — not topographic watershed divides — and each zone drains to exactly one outlet, as required for staged development and hydraulic bookkeeping. The zone polygons shown on the maps are the dissolved territories of these nodes, clipped to the built-up envelope.

Gravity feasibility was screened by accumulating pipe invert levels along every route: starting at 1.9 m below ground at the network heads, the invert falls at the Table 11 minimum gradient applicable to the accumulated upstream load (DN200 head sewers at 5.0 mm/m grading down to 0.75 mm/m for trunk-scale flows), riding no shallower than minimum cover where the ground falls faster. Wherever the accumulated depth exceeds the 12 m maximum cover of the guideline (p33), gravity is deemed infeasible for that route and the node is flagged. Contiguous flagged areas were consolidated: each becomes a single candidate sewage lifting station at its lowest point, and nearby stations within 1.5 km were cascaded into one. Pockets below 50 plots were set aside for resolution by local re-routing during design.



The method deliberately over-detects at the margins — it cannot re-grade individual streets or exploit private-land shortcuts — and its outputs are candidates for design, not designed stations. Its value is that every flagged location follows deterministically from the guideline gradients, the cover limit and the terrain, so the concept design can concentrate on eliminating stations rather than finding them.



7. CONCEPT FINDINGS

7.1. GRAVITY FEASIBILITY AND TRUNK SYSTEM

The main agglomeration commands a median available fall of approximately 3.6 m/km towards the existing STP, with the main trunk corridor descending monotonically over its 22 km length. Approximately 84% of all plots reach the STP within the guideline gradients and cover limits by gravity alone. The concept trunk system comprises 22 km of main trunk and about 172 km of trunk branches along major-road corridors.

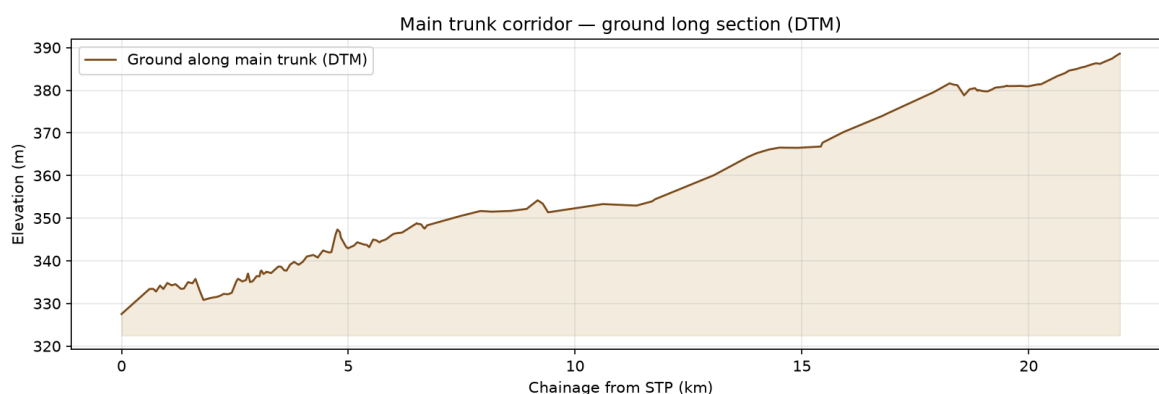
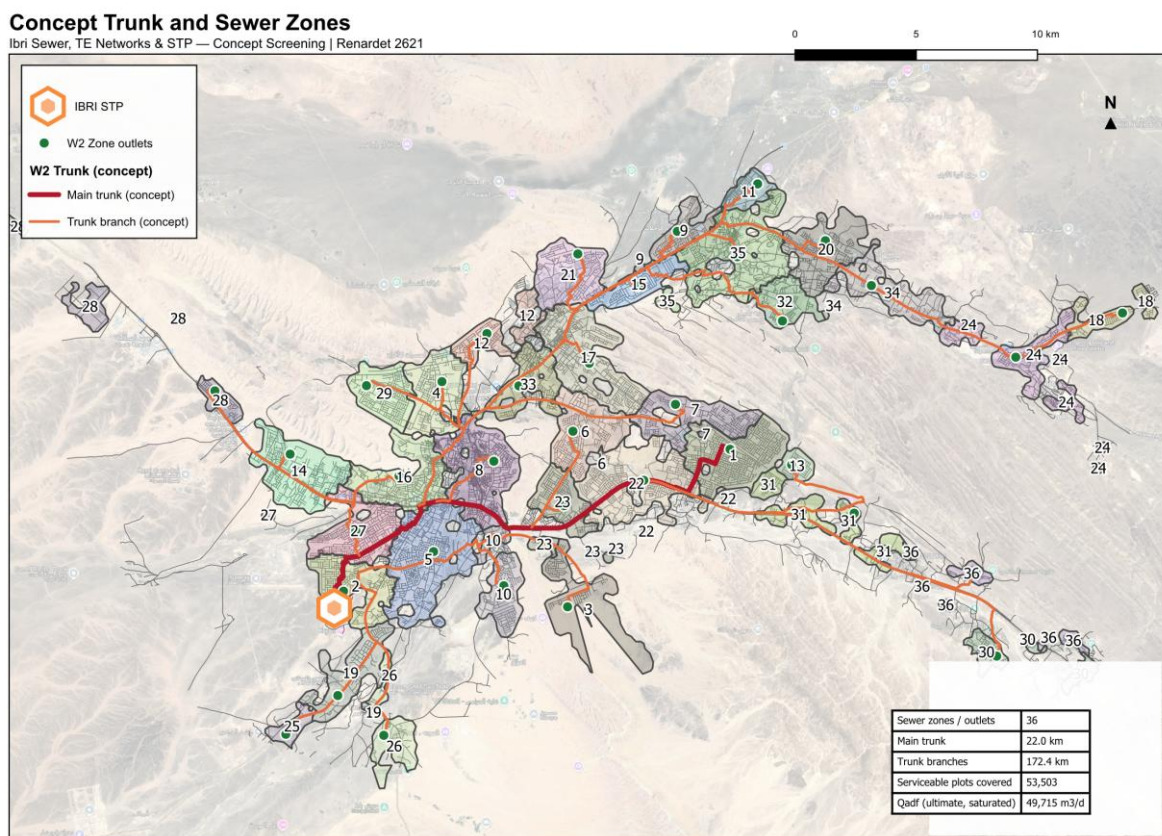


Figure — Ground long-section along the main trunk corridor (DTM)



Map M4 — Concept trunk and sewer zones

7.2. SEWER ZONES AND DESIGN FLOWS

The 36 zones and their ultimate saturated flows are listed in the Appendix. Aggregate results:

Quantity	Value
Serviceable plots (all classes)	53,503
Population at provisional occupancy (6.0 p/hu)	~279,800 (saturated)
Average daily flow Qadf (ultimate)	~49,700 m3/d
Qadf + 10% STP margin	~54,700 m3/d
Peak flow (Peltier, by zone)	~86,400 m3/d
Collection network length (screening)	~1,290 km streets carrying sewers

Table — Ultimate flow summary (provisional occupancy; scales with NCSI data)

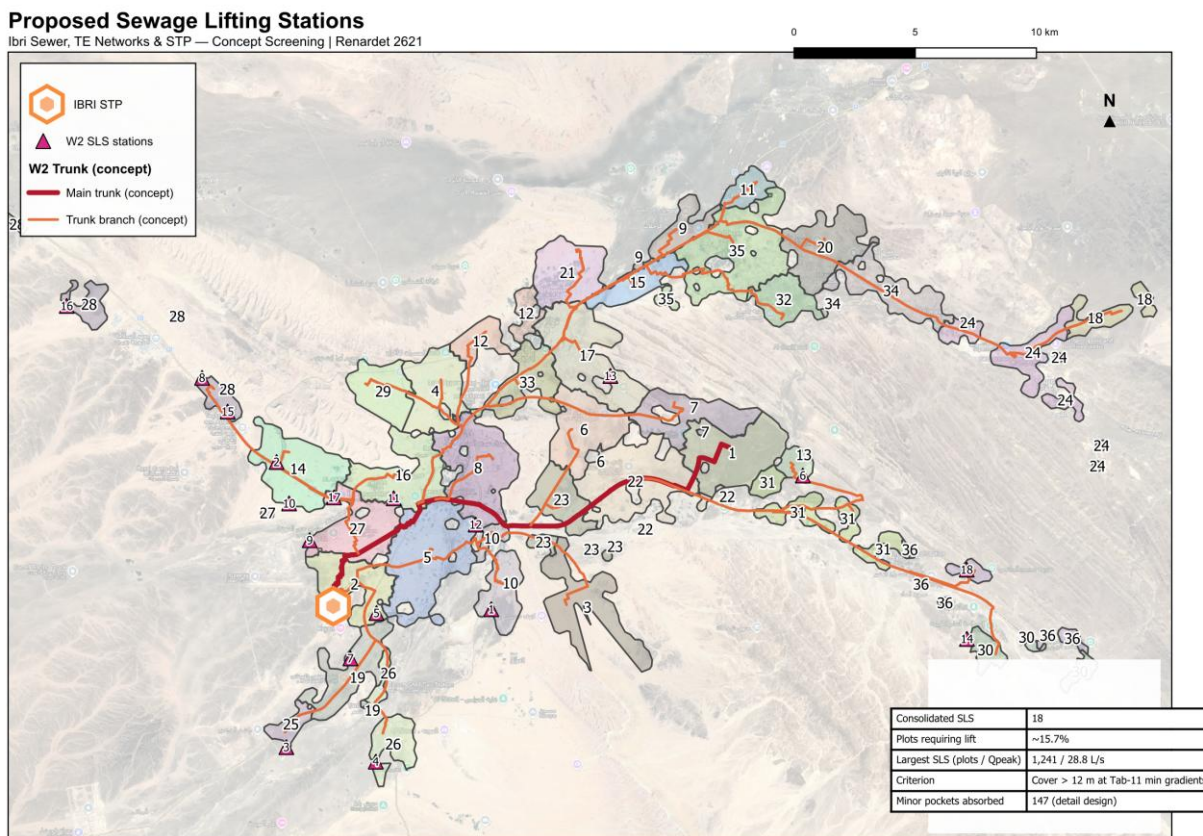
The ultimate capacity requirement is therefore roughly 2.5 times the 20,000 m3/d threshold below which the STP preliminary design and EPC tender fall automatically within this consultancy. The

immediate consequence is that STP phasing drives the commercial and technical structure of the project: Phase I sizing must follow from the 2030/2055 build-out projections — not from saturation — and those projections require the population, occupancy and development-rate data listed in Section 8. At typical land footprints of 0.9-3.6 m² per m³/d (GUD-203 Table 28), the ultimate treatment capacity implies 5-20 ha depending on technology, which should be secured at the STP site now even if construction is phased.

7.3. SEWAGE LIFTING STATIONS

Eighteen candidate SLS remain after consolidation, concentrated in wadi-floor settlements south-west and north-west of the town centre that lie below the STP inlet level or below their surrounding drainage fabric. Together they lift about 16% of the plots. The three largest candidates serve approximately 1,240, 1,070 and 890 plots with screening peak flows of 29, 26 and 19 L/s respectively. During concept design each candidate shall be tested against: (i) elimination by local re-grading or re-routing; (ii) cascading into a neighbouring station; (iii) retention with force main to the nearest gravity trunk; and for the remote eastern settlements, (iv) comparison of long conveyance against local (satellite) treatment on whole-life cost, as the options analysis requires (G1-p95-97).





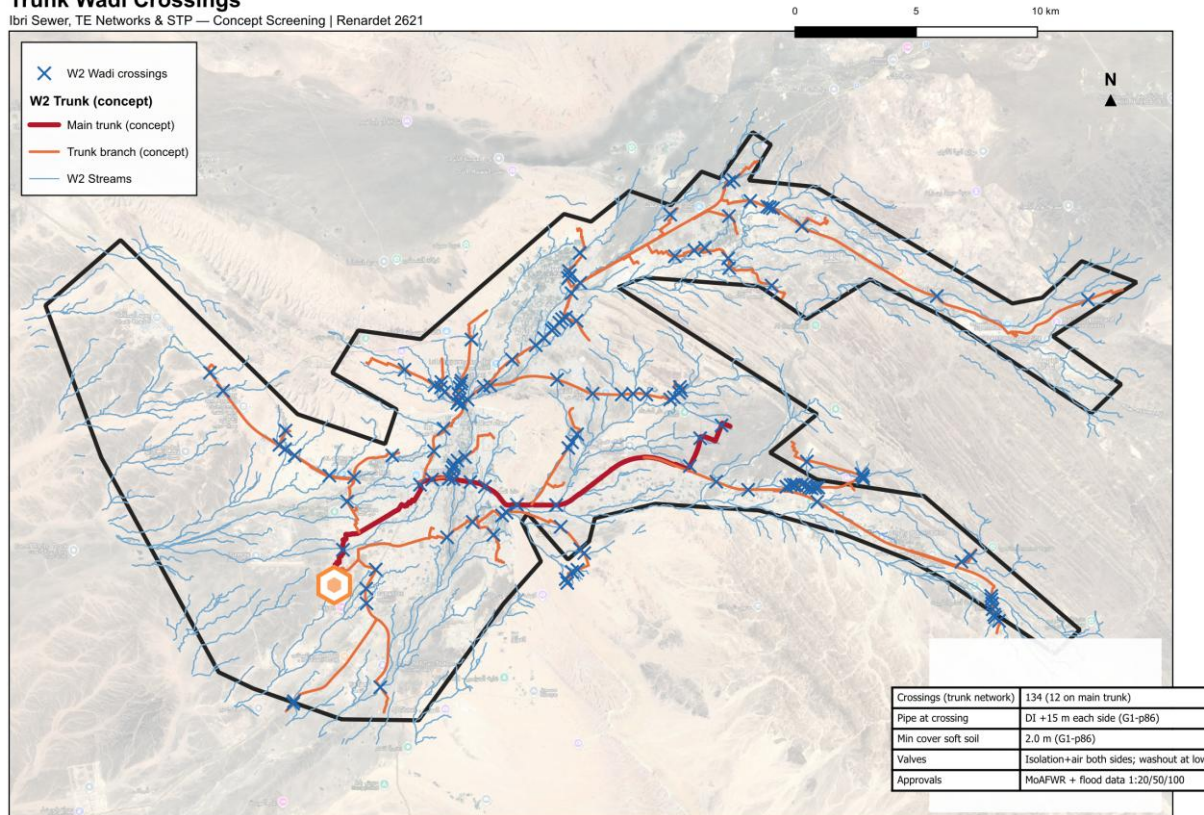
Map M5 — Proposed sewage lifting stations (candidates)

7.4. WADI CROSSINGS

The concept trunk network crosses mapped stream lines at approximately 134 locations, of which 12 lie on the main trunk. Each crossing carries the design obligations summarised in Section 4 (ductile iron across the crossing +15 m, 2.0 m cover in soft soil, valve arrangement, flood analysis and MoAFWR approval). The stream network derives from the terrain model and shall be re-based on the topographic survey; crossings will be a significant cost and approval item and early engagement with MoAFWR is recommended.

Trunk Wadi Crossings

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Map M6 — Trunk wadi crossings

8. DATA REQUIRED FROM NWS AND THIRD PARTIES

The following register lists the data needed to convert this screening into the concept design. Items marked (critical) gate the flow projections and the STP phasing decision.

#	Data item	Source	Use
1	Integrated Master Plan 2024 extracts for Ibri: population forecast, per-capita updates, phasing assumptions (critical)	NWS Planning	Flow projections 2030/2055
2	NCSI census and housing data at wilayat/settlement level: population, housing units, occupancy rate (critical)	NCSI via NWS	Population from plots
3	Water billing / consumption records (3 years) and active account counts by category; electricity account counts for pro-rata distribution	NWS	Calibrate demand and occupancy
4	Water network coverage plans and NRW figures for Ibri	NWS	Return-rate assumptions
5	Existing sewer network as-builts, GIS, connected customer list, house-connection records	NWS O&M	Integration and redesign scope
6	Existing Ibri STP: design capacity, process, headworks invert levels, spare capacity, O&M constraints (critical)	NWS	Trunk terminal level; integration
7	STP inflow records — flow meter data, daily/diurnal, minimum 12 months (critical)	NWS SCADA/O&M	Calibrate flows and peak factors
8	Influent/effluent quality records (LIMS) for existing STP	NWS LIMS	Process design basis
9	Existing lifting/pumping stations: locations, drawings, capacities, SCADA flow data	NWS	Reuse/integration assessment
10	Yellow-tanker discharge records at the STP; tanker filling station data	NWS	Present-day flow reconciliation
11	TE system: existing network as-builts, current customers and demands, TSE quality records	NWS	TE network design
12	RG Master Plan 2018 and Figures F2/F3 boundaries in GIS format	NWS	Existing vs new design areas
13	MoHUP: planned land-use changes, subdivision schemes, land bank for STP/SLS sites	MoHUP	Zone build-out; site securing
14	Utility records: water, electricity (OETC/MEDC), telecom, gas within the corridor network	Utilities	Trial pit targeting; clash analysis
15	Existing geotechnical investigations and groundwater monitoring in Ibri	NWS / MoT / others	GI planning
16	Wadi flood studies, aflaj inventory and protection requirements	MoAFWR / CAA	Crossing design and approvals
17	Survey control: benchmarks and national datum references in the wilayat	NSA / NWS	Topographic survey datum
18	NWS standard drawings (PAM-STD series), as-built/GIS specifications, BD data requirements	NWS	Deliverable compliance



19	Existing sewer & TE network files: as-built DWG/CAD and GIS/shapefiles, incl. Figure F2 source data (critical)	NWS	Integration; served-area mapping
20	Previous design reports, hydraulic models and calculations of the existing system and trunk to the STP (critical)	NWS	Capacity verification
21	Existing trunk main details: diameter, material, invert profile, condition/CCTV records, connection points	NWS O&M	Reuse vs replace decision



9. CONCLUSIONS AND NEXT STEPS

- The gravity concept is confirmed as the backbone strategy: the terrain supports gravity conveyance of the great majority of the catchment to the existing STP site, in line with the TOR preference.
- STP phasing is the decisive early question; it cannot be settled without the master plan and NCSI data of Section 8, which should be requested at kick-off.
- The 18 SLS candidates and the remote eastern settlements define the option space for the mandated three network options: maximum-gravity, consolidated-pumping, and satellite-treatment variants.
- Early engagements recommended: MoAFWR (wadi crossings and flood data), MoHUP (STP/SLS land), NWS O&M (STP records and as-builts).
- Next steps: mobilise topographic survey and data collection; georeference existing-system extents; construct the SewerGEMS model on the zone structure of this report; develop and cost the three concept options; prepare the STP location and phasing report.



10. APPENDIX A — ZONE SCHEDULE (ULTIMATE, SATURATED)

Zone	Plots	Population	Qadf (m3/d)	PF	Qpeak (m3/d)	Streets (km)
1	4762	26088	4544.2	1.64	7442.8	105.24
2	2006	10788	1893.2	1.71	3244.2	63.05
3	1539	8058	1399.8	1.75	2447.4	27.22
4	2008	10686	1875.3	1.71	3215.5	62.51
5	2879	15618	2768.5	1.68	4641.9	129.78
6	2836	13050	2281.8	1.69	3866.8	64.74
7	2299	12756	2222.8	1.7	3772.5	52.7
8	2466	12720	2289.2	1.69	3878.5	153.45
9	933	4860	861.7	1.82	1565.5	40.7
10	1348	6828	1219.8	1.77	2154.4	69.88
11	670	3636	639.5	1.87	1194.4	23.28
12	901	4938	879.9	1.81	1595.5	47.32
13	540	2910	508.4	1.91	972.2	13.87
14	2252	10836	1937.5	1.71	3315.3	113.13
15	982	5310	943.2	1.8	1700.3	46.8
16	1750	9054	1618.4	1.73	2801.5	93.91
17	2782	14070	2490.8	1.69	4200.0	112.25
18	614	2934	519.8	1.91	991.6	23.97
19	1206	4776	854.6	1.82	1553.6	50.73
20	1580	8820	1550.2	1.74	2691.3	54.9
21	1304	6738	1203.2	1.77	2127.2	68.19
22	2885	15234	2706.8	1.68	4543.8	135.44
23	1338	7230	1291.7	1.76	2271.7	74.11
24	859	4506	836.0	1.82	1522.7	89.14
25	451	2502	440.5	1.94	855.9	16.63
26	823	4614	820.9	1.82	1497.7	42.51
27	2018	10554	1874.6	1.71	3214.4	92.92
28	556	2766	515.8	1.91	984.8	58.37
29	1746	9180	1610.4	1.73	2788.6	52.84
30	372	1872	357.3	1.99	711.6	50.85
31	942	5124	953.4	1.8	1717.2	105.26
32	541	3018	539.1	1.9	1024.4	30.75
33	880	4680	855.6	1.82	1555.3	74.98
34	599	3300	602.0	1.88	1131.1	51.1
35	1597	8334	1524.6	1.74	2649.9	134.96
36	239	1368	284.3	2.05	583.2	69.43

Population at provisional occupancy 6.0 persons/housing unit; to be updated with NCSI data.

